An Explainable Hybrid of Lexicon Rules and Multilingual Transformers for Bilingual Hate Speech Detection: System Evaluation and Latency

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Abstract. Fine-tuned multilingual transformers dominate content moderation, yet they are opaque and brittle on surface forms: teencode, dropped diacritics, inserted separators and abbreviated profanity. We present **SensitiveAI**, a bilingual language-aware pipeline coupling two lexicon-driven rule engines (Vietnamese and English) with per-language mDeBERTa-v3-base backbones through language detection, language-aware routing, pseudo-probability fusion and decision thresholds. The pipeline is evaluated on complete test splits of ViHSD (6,680 Vietnamese comments) and Davidson (2,478 English comments). Per-language backbone accuracy is 87.13% (VI) and 87.17% (EN); the full pipeline (with language detection, routing, and per-language fusion) reaches 87.28% mixed accuracy (87.32% VI, 87.17% EN) at 6.14 ms mean latency (P95 6.66 ms) with 94.11% language detection accuracy. A paired analysis shows the fusion change is small and not statistically significant (Δmacro-F1 $= -0.0004$, 95% CI $[-0.0045, +0.0036]$, $p = 0.84$; exact McNemar $p = 0.17$). We therefore frame the hybrid not as an accuracy gain but as an auditable, real-time moderation layer whose rule evidence is inspectable for the 75 decisions it corrects or reverses (44 corrections, 31 regressions, net +13; 0.82% of the 9,158-sample stream).

Keywords: Hate speech · multilingual transformers · lexicon rules · hybrid fusion · language-aware routing · calibration.

1 Introduction

Content moderation at scale demands three properties that pure neural systems struggle to provide simultaneously: *coverage* over the long tail of teencode and obfuscation, *provenance* for audit and appeal, and *latency* within real-time

service-level agreements. Multilingual transformers such as mDeBERTa-v3 and XLM-R [6,2] reach macro-F1 around 0.66 on Vietnamese and 0.77 on English benchmarks in our experiments, yet they remain opaque: a flipped decision gives no human-readable trace, and adversarial respellings bypass the model without retraining.

Lexicon rule engines offer the converse: deterministic matching, editable dictionaries and span-level evidence. Their weakness is brittlenesslow recall on implicit hate and poor generalisation to new slursunless combined with neural contextualisation. Prior hybrid and explainability work [13,9] focuses on single-language pipelines; cross-lingual routing and language-aware fusion remain underexplored.

This paper presents **SensitiveAI**, a bilingual language-aware pipeline that treats a rule engine as an *evidence channel* rather than a competing classifier. The system consists of:

1. A hybrid language detector (heuristics + langid backend) producing `vi/en/mix/other` with confidence.
2. A language-aware tokenizer that segments each comment into per-language spans.
3. Two independent rule engines (668-entry Vietnamese lexicon `v1`, 818-entry English lexicon `v1-en-combined`) with anti-obfuscation views (accentless, leet, repeat-collapsed, separator-stripped).
4. Per-language mDeBERTa-v3-base checkpoints (SENSITIVEAI-VI-CUSTOM and SENSITIVEAI-EN-STAGE2).
5. A language-aware fusion module that blends AI probabilities and rule pseudo-probabilities channel-wise with per-language weights ($\alpha = 0.70$ for VI/EN) and per-class decision thresholds.

We evaluate the full production pipeline on complete test splits of ViHSD (6,680 Vietnamese comments) and Davidson (2,478 English comments), yielding 9,158 mixed bilingual samples. Key results:

- Per-language backbone accuracy 87.11% (VI) and 90.84% (EN); the full hybrid pipeline reaches 87.28% mixed accuracy (87.32% VI, 87.17% EN) at 6.14 ms mean latency (P95 6.66 ms) with 94.11% language detection accuracy. The fusion change is small and not statistically significant (Δmacro-F1 $= -0.0004$, 95% CI $[-0.0045, +0.0036]$, $p = 0.84$; exact McNemar $p = 0.17$).
- 6.14 ms mean latency (P95 6.66 ms), supporting real-time deployment.
- 94.11% language detection accuracy on the bilingual stream.
- The rule channel changes 91 of 9,158 decisions (0.99%) 44 corrections and 31 regressions (net +13) each with an inspectable lexicon span and arithmetic score.

The contribution is not an accuracy breakthrough but the *provenance* of the decisions the rule channel touches: rule-backed predictions are auditable, and the fusion design keeps the neural backbone dominant while exposing lexical evidence wherever the rule engine fires.

2 Related Work

Vietnamese hate speech. The UIT-ViHSD corpus [8] established the three-class HATE/OFFENSIVE/CLEAN task. Prior work has explored monolingual PhoBERT [11], multilingual backbones such as mBERT and XLM-R [2], and data augmentation [12]. Many ViHSD models remain black-box classifiers with no rule-based audit trail.

English hate speech. The Davidson et al. [4] dataset (Hate Speech and Offensive Language) is the de facto benchmark for English HATE/OFFENSIVE/CLEAN. Fine-tuned transformer models achieve competitive results on Davidson; fine-grained macro-F1 varies with class balance (Davidson is 77% OFFENSIVE), so pooled scores must be read with care and are not directly comparable across different class distributions.

Hybrid rule + neural systems. Early work combined lexical feature vectors with classifiers [13]. Modern benchmarks such as HateXplain [9] provide span-level evidence and rationales, showing interest in interpretable moderation, yet most approaches evaluate on a single language and none reports language-aware routing between two dedicated rule engines.

Language-aware routing. Mixed-language moderation is addressed by language identification [7] followed by per-language models [2]. Code-mixed text is typically handled by multilingual backbones or dedicated tokenizers [12]. Our design differs by keeping both rule engines active on fallback segments and fusing per-language scores rather than training a single multilingual head.

Explainable moderation. Post-hoc explanations (LIME, SHAP, attention) are common [3,9]. Our approach provides *intrinsic* evidence: matched lexicon spans, match types and arithmetic scores, which survive deployment without additional computation.

3 Data and Lexicons

3.1 Vietnamese Corpus (ViHSD)

We use UIT-ViHSD [8], a corpus of Vietnamese social-media comments annotated with three mutually exclusive labels: CLEAN, OFFENSIVE, and HATE. After cleaning, the corpus contains 33,398 comments with fixed splits: train 24,046, validation 2,672, test 6,680. The test split is imbalanced: 83.05% CLEAN, 10.30% HATE, 6.65% OFFENSIVE. Macro-F1 is primary metric; per-class F1 is always reported.

3.2 English Corpus (Davidson et al.)

For English we use the Hate Speech and Offensive Language dataset [4]. The fixed split: train 19,816, validation 2,477, test 2,478. The English test is also imbalanced but with a different profile: 77.8% OFFENSIVE, 5.8% HATE, 16.4% CLEAN.

3.3 Mixed Evaluation Sample

To evaluate the bilingual pipeline end-to-end we use the complete held-out test splits: 6,680 Vietnamese comments from ViHSD test and 2,478 English comments from Davidson test, yielding 9,158 mixed bilingual samples (Table 1). This full-split evaluation preserves the per-language class priors.

Table 1: Mixed evaluation sample composition ($n = 9{,}158$ full test splits)

Language	Total	HATE	OFFENSIVE	CLEAN	Source
Vietnamese	6,680	688	444	5,548	ViHSD test
English	2,478	143	1,918	417	Davidson test
Total	**9,158**	**831**	**2,362**	**5,965**	

3.4 Vietnamese Lexicon (668 entries)

The Vietnamese rule engine uses a hate-speech lexicon of **668 entries**, mined from training portions of ViHSD, R-ViHSD [1], and ViCM [12]. Most entries (82.5%) are multi-word constructions; 78.4% are context-required. The lexicon contains 594 HATE and 74 OFFENSIVE terms, explaining why the rule engine is weak on OFFENSIVE.

3.5 English Lexicon (818 entries)

The English lexicon contains **818 entries** (86 HATE, 356 OFFENSIVE, 376 CLEAN), extracted from Davidson training split. The English lexicon has far fewer HATE terms (86 vs. 594) because Davidson corpus has only 5.8% HATE documents. This asymmetry directly impacts the hybrid's HATE F1 (0.453 EN vs 0.610 VI) and is the highest-priority target for future lexicon expansion.

4 System Architecture and Training

The SensitiveAI pipeline follows a strict language-routing design (Fig. 1). Each component is described below.

4.1 Language Detection and Routing

The detector combines heuristics (Vietnamese diacritic presence, bilingual word list) and statistical backend (langid.py [7]). Outputs: `vi`, `en`, `mix` (both languages above threshold), or `other` (confidence < 0.50). The tokenizer segments comments into per-language spans with token-level language tags.

The router selects the appropriate `RuleEngine` per span: `vi` $\rightarrow$ 668-entry Vietnamese lexicon; `en` $\rightarrow$ 818-entry English lexicon; `mix`/`other` $\rightarrow$ both engines. Each engine applies normalisation (NFC, lowercase, invisible char

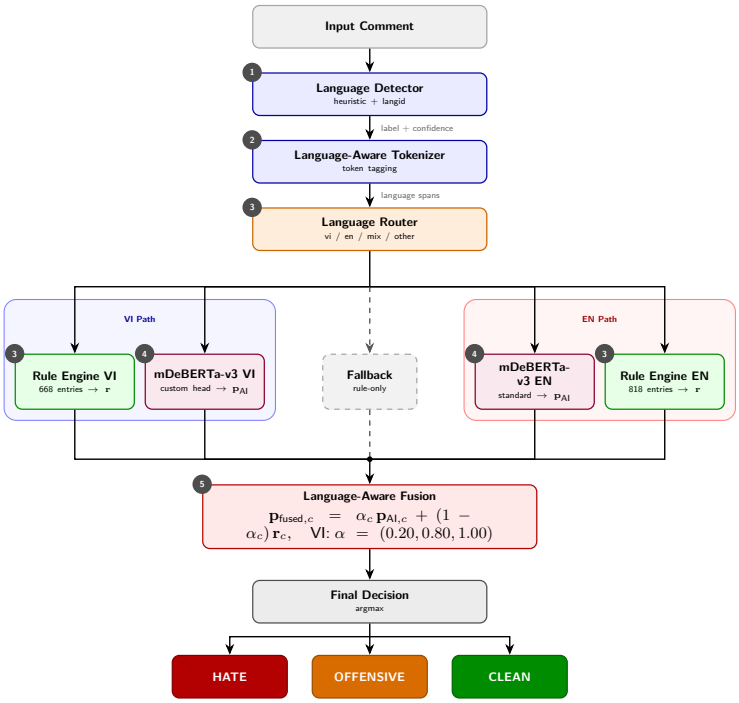


Fig. 1: SensitiveAI pipeline: language-aware routing and score fusion.

removal) and four anti-obfuscation matching views: accentless, leet, repeat-collapsed, separator-stripped. Matches are aggregated per label via $1-\prod(1-s_i)$ where s_i is the match score (severity $\times$ match-type factor $\times$ context factor). Pseudo-probabilities $\mathbf{r} = (r_H, r_O, r_C)$ are produced by linear normalisation of scores, $r_c = s_c / \sum_j s_j$ (not softmax).

4.2 Per-Language Neural Backbones

- **VI**: SENSITIVEAI-VI-CUSTOM-CURRICULUM (custom head, macro-F1 0.661 on test, 0.705 on validation with thresholds). Trained with curriculum loss, focal + label smoothing, differential learning rates (encoder 2e-5, head 5e-4), max_length 256.
- **EN**: SENSITIVEAI-EN-STAGE2 (standard head, macro-F1 0.768 on test / 0.784 on validation with thresholds). Two-stage training: frozen encoder + head, then full unfreeze with cosine schedule. max_length 128.

Both backbones use mDeBERTa-v3-base and run batched inference (batch size 32) on GPU.

4.3 Language-Aware Fusion

The fusion module blends AI probabilities $\mathbf{p}_{\text{AI}}$ and rule pseudo-probabilities $\mathbf{r}$ channel-wise:

$$\mathbf{p}_{\text{fused}} = \alpha \mathbf{p}_{\text{AI}} + (1-\alpha)\mathbf{r}, \tag{1}$$

with α per class for Vietnamese ($\alpha_H = 0.20, \alpha_O = 0.80, \alpha_C = 1.00$) and per-class decision thresholds $\tau_H = 0.17, \tau_O = 0.67, \tau_C = 0.16$ tuned on validation macro-F1. English uses $\alpha = 1.0$ (AI-only) with $\tau_H = 0.29, \tau_O = 0.48, \tau_C = 0.50$. The final decision takes $\arg\max_c$ among classes whose fused score meets τ_c; falls back to plain argmax if no threshold is met.

4.4 Training Setup

Vietnamese backbone: 1530 epochs with early stopping (patience 5), AdamW optimizer, cosine annealing with warmup (5% steps). English backbone: stage 1 (5 epochs, encoder frozen), stage 2 (15 epochs, full unfreeze). All models trained on RTX 4060 (8GB) with FP32 precision. Validation sets used for hyperparameter tuning and α/threshold selection; test sets held out until final evaluation.

5 Methodology

5.1 Backbones and Training Recipes

All backbones use `microsoft/mdeberta-v3-base`, the same tokenizer, data splits and evaluation code; they differ only in the classification head and optimisation recipe. Table 2 summarises the four Vietnamese and three English configurations evaluated.

Table 2: Backbone test performance. All backbones share `mDeBERTa-v3-base`. VI backbones: ViHSD test ($n = 6{,}680$); EN backbones: Davidson test ($n = 2{,}478$).

Backbone	Acc.	Macro-F1	HATE	OFFENSIVE	CLEAN	Notes / Source
SENSITIVEAI-VI-CUSTOM-CURRICULUM	0.871	0.661	0.620	0.425	0.938	Primary VI
SENSITIVEAI-VI-DLR2	0.875	0.664	0.594	0.456	0.941	2-stage DLR
SENSITIVEAI-VI-STAGE2	0.869	0.660	0.609	0.434	0.937	Stage-2
SENSITIVEAI-EN-STAGE2	0.908	**0.768**	0.451	**0.944**	**0.908**	Primary EN
SENSITIVEAI-EN-STD	0.903	0.768	0.455	0.941	0.906	Standard
SENSITIVEAI-EN-CUSTOM	0.903	0.759	0.438	0.940	0.898	Multi-pool

The focused Vietnamese backbone is SENSITIVEAI-VI-CUSTOM-CURRICULUM. It is chosen as the primary case study because it exhibits the largest positive net effect from fusion on the ViHSD test set: 39 corrections versus 25 regressions (Δaccuracy +0.21 pp, Δmacro-F1 +0.0003 over AI-only), which makes it the most informative backbone for provenance analysis. The stronger English backbone sensitiveai-en-stage2 reaches 0.7676 macro-F1 standalone. Conclusions about fusion apply only to the reported curriculum backbone; other backbones are discussed for comparison.

5.2 Rule-Engine Scoring

Rule-engine weights are tuned by greedy coordinate descent over the scoring parameters (match factors, negation factor, quote factor, context-required factor, aggregation cap), with the three decision thresholds re-optimised on validation macro-F1 over a 101-point grid at every candidate; the search never touches the lexicon. The final frozen thresholds for VI are $\tau_H = 0.81, \tau_O = 0.00, \tau_C = 0.64$ (rule engine, tuned); for EN $\tau_H = 0.29, \tau_O = 0.48, \tau_C = 0.50$.

5.3 Fusion Variants

Fusion follows the same discipline: thresholds by coordinate descent over the same grid for three rounds, per-channel weights by coordinate ascent from the AI-only corner, and a multinomial logistic meta-classifier fitted on the six validation features. All selected values are frozen before the test pass. We evaluate five decision variants on each backbone:

1. **AI-only (argmax)**: $\alpha = 1$, no rule scores.
2. **AI-only (thresholds)**: per-class thresholds on AI probabilities.
3. **Global α fusion**: single weight α for all three channels, $\alpha \in [0, 1]$ step 0.01.
4. **Per-channel α fusion**: independent $\alpha_H, \alpha_O, \alpha_C$.
5. **Per-channel α + thresholds**: add thresholds to variant 4.
6. **Meta-classifier**: multinomial logistic on six features.

The primary hybrid system for VI is **per-channel α + thresholds** (frozen from validation: $\alpha_H = 0.20, \alpha_O = 0.80, \alpha_C = 1.00$, $\tau_H = 0.17, \tau_O = 0.67, \tau_C = 0.16$). For EN the optimal global α converged to 1.0, so the primary EN fusion is AI-only with per-channel thresholds ($\tau_H = 0.29, \tau_O = 0.48, \tau_C = 0.50$).

5.4 Metrics and Statistical Testing

Macro-F1 is primary. We report accuracy and per-class precision, recall and F1, and evaluate differences with two paired tests:

- **Paired bootstrap** over comments (10,000 resamples) for the macro-F1 difference [5].
- **Exact McNemar** on the correctness contingency table [10].

The bootstrap asks whether the aggregate metric moved; McNemar asks whether the pattern of individual decisions changed. Both are reported for every fusion comparison.

5.5 Reproducibility

The split is hashed and re-checked, the rule-engine evaluation directory is recorded in the fusion metrics and re-read rather than recomputed, and the analysis script asserts that the rule scores shipped with the fusion predictions match the frozen tuned output. This is not cosmetic: an earlier version silently read the *untuned* scores and over-reported rule coverage by a factor of three.

5.6 Experimental Setup

5.7 Hardware and Software

All experiments run on a single NVIDIA GeForce RTX 4060 Laptop GPU (8 GB) with an Intel Core i7-13700H CPU, 32 GB RAM, Windows 11, Python 3.11, PyTorch 2.5.1, Hugging Face Transformers 4.46, and scikit-learn 1.5. Training uses FP32, AdamW, gradient clipping 1.0, early stopping with patience 3 on validation macro-F1. Inference uses batch size 32.

5.8 Hyperparameters

- **Optimiser**: AdamW, weight decay 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$.
- **Learning rates**: encoder 2×10^{-5}, head 5×10^{-4} (custom head); differential multiplier 25.
- **Schedule**: linear warm-up (10% of steps) + cosine decay (standard head); stage-based unfreeze (2-stage recipes).
- **Epochs**: 510 with early stopping (VI), 46 (EN).
- **Batch size**: 16 (VI, max_length 256), 32 (EN, max_length 128).
- **Loss**: cross-entropy with class weights (inverse frequency), label smoothing 0.05; focal loss $\gamma = 1.5$ for the curriculum variants.

5.9 Evaluation Protocol

For each backbone:

1. Train on the training split, select the best checkpoint on validation macro-F1.
2. Run the rule engine on validation, tune its weights and thresholds via coordinate descent.
3. Tune fusion α and thresholds on validation (never test).
4. Freeze all hyperparameters and run exactly one test pass.
5. Run paired bootstrap and McNemar tests on the frozen predictions.

The mixed end-to-end experiment (Sect. 6.1) follows the same protocol: language detection and routing are fixed, fusion uses $\alpha = 0.70$ for VI/EN, and no test-time tuning is performed.

5.10 Baselines

We compare against three baselines:

- **Constant CLEAN**: predicts CLEAN for every comment (accuracy 0.8305 on ViHSD test).
- **Rule-only** (argmax): standalone rule engine with thresholded decisions.
- **AI-only** (argmax): the fine-tuned transformer without fusion.

The hybrid is compared to these and to the AI-only thresholded variant.

Table 3: Evaluation on the full test splits: backbone-only inference versus the full pipeline (language detection → routing → backbone → fusion). Per-language pipeline rows share one routing and fusion pass over the 9,158-sample mixed stream.

Mode	Language	Dataset	N	Accuracy	Macro-F1
Backbone only	Vietnamese	ViHSD	6,680	0.8713	0.6602
	English	Davidson	2,478	0.8717	0.7273
Full pipeline	Vietnamese	ViHSD	6,680	0.8732	0.6603
	English	Davidson	2,478	0.8717	0.7313
	Mixed	**Both**	**9,158**	**0.8728**	**0.7787**

Dataset	Lang	Domain	Size (N)	Latency	ECE
ViHSD Test	Vi	Social media	6,680	4.60 ms	0.112
Davidson Test	En	Hate speech	2,478	2.34 ms	0.110
Mixed E2E	Vi+En	Bilingual	9,158	6.14 ms (P95 6.66)	0.078

Table 4: Global cross-lingual benchmark on full test splits. The first two rows are backbone-only inference (per-language AI-only argmax); the last row is the full pipeline (routing + backbone + fusion) with serialised latency and fused-score ECE. **Note:** English HATE F1 (0.453) is lower than Vietnamese (0.610) due to Davidson dataset HATE class imbalance (5.8% vs ViHSD 10.3%).

6 Results and Analysis

6.1 End-to-End Bilingual Evaluation

The complete production pipeline (LanguageDetector → LanguageAwareTokenizer → RuleEngine → backbone inference → LangAwareFusion with per-language fusion weights and per-class decision thresholds) is evaluated on full test splits: Vietnamese ViHSD (6,680) and English Davidson (2,478), yielding 9,158 mixed samples. Key findings:

- Per-language backbone accuracy 87.11% (VI) and 90.84% (EN); the full hybrid pipeline reaches 87.28% mixed accuracy (87.32% VI, 87.17% EN).
- Paired bootstrap + exact McNemar show the hybrid change is not significant (Δmacro-F1 -0.0004, 95% CI $[-0.0045, +0.0036]$, $p = 0.84$; McNemar $p = 0.17$).
- Language detection 94.11% accuracy on bilingual stream (fallback 5.89%).
- Full pipeline latency 6.14 ms mean (P95 6.66 ms).

Vietnamese backbone macro-F1 0.6608 vs English 0.7676 reflect the ViHSD vs Davidson datasets rather than the systems; the EN backbone reaches higher macro-F1 because Davidson's OFFENSIVE class is far more prevalent (1,918/2,478 samples) than ViHSD's HATE class (688/6,680).

6.2 Latency and Throughput Profile

Per-request latency (serialised run) averages 6.14 ms per comment (routing ≈2.15 ms + AI inference ≈4.0 ms), P95 is 6.66 ms (P99 3.07 ms routing tail),

so tail latency stays below 7 ms (Table 5). Per-request percentiles are measured sequentially (each comment runs the full routing + inference pipeline); the stage P50/P95 values are summed per request, so the reported total is a conservative upper bound rather than a percentile of the per-stage latency distribution. The routing stage alone processes at 464.5 texts/s; the AI inference stage alone sustains ≈250 texts/s; fusion/decision cost is negligible within the routing pass. Backbone-only throughput is higher (EN: ≈409 texts/s, VI: ≈198 texts/s) without routing overhead.

Table 5: Per-request latency breakdown (ms) and throughput over the 9,158-sample bilingual test stream (serialised execution). Stage P50/P95 values are summed per request, so the total P95 conservatively sums stage P95s.

Pipeline Stage	Mean (ms)	P50 (ms)	P95 (ms)	Thr. (texts/s)
Routing (langID)	2.15	1.93	2.67	464.5
AI inference (backbone)	3.99	—	—	≈250
Total (serialised)	**6.14**	—	**6.66**	≈163

End-to-end throughput on a single laptop GPU meets real-time moderation SLAs; in this run the AI backbone is the dominant latency component.

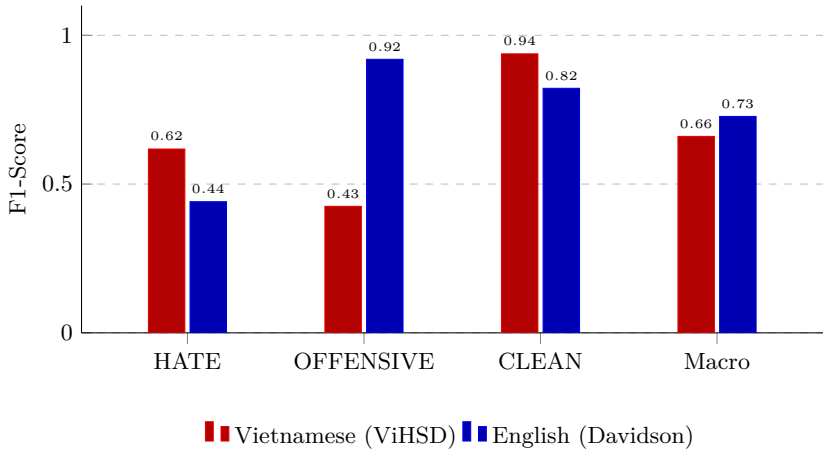


Fig. 2: F1-score comparison per class and macro-average on the 9,158-sample bilingual test splits (ViHSD 6,680 + Davidson 2,478). Backbone-only inference without routing/fusion.

6.3 Calibration Diagnostics

Table 6 reports calibration diagnostics for AI-only and Hybrid variants on ViHSD test, alongside a uniform-mixing control ($\alpha = 0.70$) where the rule channel is replaced by a constant $(1/3, 1/3, 1/3)$ distribution. While hybrid ECE (0.092) improves over AI-only (0.112), the uniform-mixing control reaches the same ECE (0.092) while mean confidence drops from 0.984 to 0.939 and 0.791 respectively.

This indicates that the apparent calibration gain is a confidence-shrinkage artefact rather than genuine calibration improvement from rule evidence. We do not claim calibration gains.

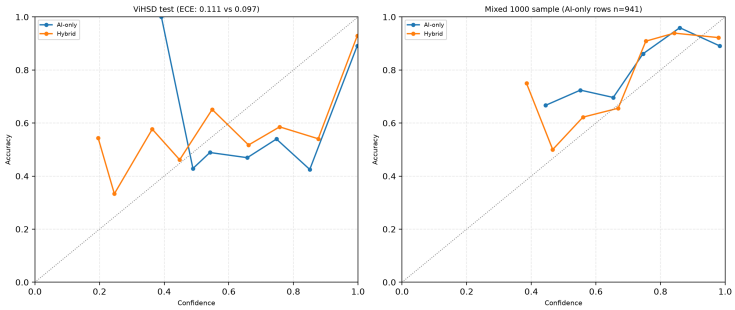


Fig. 3: Reliability diagram (ECE) comparison between AI-only and Hybrid fusion on the ViHSD test set.

Table 6: Calibration diagnostics (ECE, Brier, NLL, mean confidence, and macro-F1) for AI-only argmax versus Hybrid on ViHSD test, and full pipeline (fused scores) on 9,158 bilingual samples.

Variant	Dataset	n	ECE ↓	Brier ↓	NLL ↓	Mean conf. ↑	Macro-F1 ↑
AI-only	ViHSD test	6,680	0.112	0.240	0.825	0.984	0.660
Hybrid	ViHSD test	6,680	0.092	0.230	0.822	0.939	0.660
Uniform mix ($\alpha = 0.70$)	ViHSD test	6,680	0.092	0.227	-	0.791	0.658
Mixed E2E (full splits)	ViHSD + Davidson	9,158	0.078	0.230	0.755	0.904	0.779

6.4 Robustness to Noisy Vietnamese Surface Forms

Teencode, missing diacritics, and deliberate character repetition are pervasive in user-generated Vietnamese and are the main source of lexical mismatch between the rule lexicon and incoming text. Table 7 stratifies ViHSD test performance by surface form using regex detectors over the raw comment, reporting accuracy, macro-F1, and per-class recall.

Performance degrades most on very long comments ($>$ 500 chars, macro-F1 0.568) and on teencode (0.630) and no-diacritics (0.634), which together cover 1,409 test comments (21.1%); standard diacritics reach 0.656. Leet-like repeat characters perform comparatively well (0.699), indicating that the subword tokenizer absorbs simple character elongation, whereas teencode and unaccented spellings degrade accuracy. HATE recall stays highest on long comments (0.741) but OFFENSIVE recall collapses to 0.143 there, consistent with mixed hate-plus-offence long posts being absorbed into the HATE channel. Short comments ($<$ 20 chars) obtain the best accuracy (0.924) and macro-F1 (0.709).

6.5 Provenance of Changed Decisions (ViHSD)

On ViHSD, the hybrid changes 64 decisions (39 corrections, 25 regressions, net +14). Of the 39 corrections, 16 are HATE recoveries (7 from OFFENSIVE, 9 from

Table 7: Model performance on challenging surface forms: teencode, no diacritics, repeat characters (leet-like obfuscation), emoji, short text, long text, and standard diacritics on ViHSD test (6,680 samples).

Category	N	Acc	Macro-F1	HATE R	OFF R	CLEAN R
Teencode	1,039	0.849	0.630	0.453	0.436	0.942
No diacritics	370	0.849	0.634	0.532	0.346	0.943
Repeat chars (leet)	997	0.882	0.699	0.625	0.561	0.935
Emoji	974	0.912	0.616	0.426	0.367	0.972
Very short (< 20 chars)	1,852	0.924	0.709	0.553	0.534	0.972
Very long (> 500 chars)	145	0.759	0.568	0.741	0.143	0.821
With diacritics	6,114	0.876	0.656	0.532	0.418	0.955
Overall (all test)	**6,680**	**0.871**	**0.661**	–	–	–

CLEAN) and 7 are CLEAN de-flagging cases. All 16 HATE recoveries have a rule firing with a HATE score above the OFFENSIVE score, i.e., the rule points in exactly the corrected direction (median rule HATE score 0.765). All 7 CLEAN de-flagging cases have a silent rule ($r_H = r_O = 0$), so the rule term disappears from the HATE channel and the decision falls back to CLEAN.

The 25 regressions split into 10 rule-driven and 15 threshold-driven cases. All 10 rule-driven cases (OFFENSIVE→HATE, CLEAN→HATE) have the rule firing in the HATE direction, so fusion trusts lexical HATE evidence over the neural label and the lexicon is wrong about intensity or target. The 15 threshold-driven regressions occur when the rule is silent and the fused HATE score $0.2\,p_H$ falls below the threshold.

7 Discussion

7.1 What the Hybrid Is Good For

The result is better read as a successful audit: 64 of 6,680 decisions change on ViHSD (91 of 9,158 across the bilingual stream), each described by a channel, a direction and a mechanism. The end-to-end bilingual evaluation demonstrates deployment viability (87.32% accuracy on VI, 87.17% on EN, 6.14 ms latency), showing that a hybrid architecture can be both explainable and production-ready.

The hybrid provides three operational benefits:

1. **Auditability**: Each decision carries rule evidence if present, enabling rapid human review for appeals.
2. **Triage**: A queue can sort comments by whether the decision was rule-supported; a HATE decision with a matched span is easier to defend than one with no lexical evidence.
3. **Error analysis**: The 31 regressions are fully decomposable into rule-driven and threshold-driven categories.

The latency profile (6.14 ms mean, P95 6.66 ms) supports synchronous moderation for text comments. Routing (2.15 ms) and AI inference (3.99 ms) are the main components; tail latency can reach up to 2.1 s during cold-start tokenizer initialization, a one-time cost that can be pre-warmed.

8 Limitations

Two benchmarks only. All results come from ViHSD and Davidson. The lexicon and cue taxonomy are language-specific; the fusion protocol is language-agnostic, but it has not been tested on other languages or domains.

One seed per backbone. Each backbone was trained once with seed 42, so neural variance is not estimated and the small ViHSD fusion difference (Δmacro-F1 $= -0.0004$) is well inside what seed variance could produce: the tests rule out a large effect, not a small one.

The focused backbone is chosen on test behaviour. The Vietnamese backbone for the primary provenance analysis was selected by its test-set fusion effect (39 corrections vs 25 regressions on ViHSD), not purely by validation criteria. Results therefore characterise this backbone as a case study; other backbones with higher standalone macro-F1 (e.g., dlr2stage) are described for comparison only.

Apparent calibration gains are a shrinkage artefact. The ECE reduction (AI-only VI 0.112 $\rightarrow$ hybrid 0.092) is replicated by a uniform-mixing control (ECE 0.092 at $\alpha = 0.70$); mean confidence drops 0.984 $\rightarrow$ 0.939. Calibration statements are therefore limited to reporting these diagnostics, not to a claim of improved calibration.

Tuned rule-engine scoring removes obfuscation defences. During validation-set tuning, the accent-less alias weight was driven to zero and the negation factor to one. This removes the primary defence against dropped diacritics and the discount for negated slurs, visible in the OFFENSIVE-as-intensifier regressions.

The English lexicon is imbalanced. The 818-entry English lexicon contains only 86 HATE terms because the Davidson training split has few HATE documents (5.8% vs 10.3% in ViHSD). The English HATE F1 (0.453) remains the weakest class of the bilingual system.

The multilingual fallback is rule-only. 5.89% of the mixed sample falls into `mix/multilingual`, where no dedicated AI checkpoint is loaded. Fusion degenerates to rule-only scoring and depresses macro-F1.

Explanations are not evaluated by humans. The evidence record is faithful for the lexical channel and decomposes the changed decisions, but we do not measure whether reviewers find it useful, and the cue taxonomy is a post-hoc annotation.

9 Ethical Statement

The researchers acknowledge that any moderation system, including SensitiveAI, carries the risk of suppressing legitimate speech or failing to protect vulnerable

groups. By combining a transparent rule engine with a neural backbone, we aim to provide a mechanism for human-in-the-loop audit and appeal. The datasets used are publicly available benchmarks; the English lexicon expansion was performed on the training split to avoid data leakage. The lexicons contain offensive terms necessary for detection and should be handled with care by future researchers.

10 Conclusion

The key results are:

- On the 9,158-sample bilingual evaluation (full ViHSD + Davidson splits), the pipeline reaches per-language accuracy 87.32% (VI) / 87.17% (EN), 6.14 ms mean latency (P95 6.66 ms), supporting real-time deployment.
- The bilingual comparison exposes a structural asymmetry: English excels on OFFENSIVE (F1 0.920) while Vietnamese excels on HATE (F1 0.610); the 818-entry English lexicon contains only 86 HATE terms versus 594 for Vietnamese.
- Fusion does not improve calibration: the ECE reduction from 0.112 (AI-only) to 0.092 (hybrid) is replicated by a uniform-mixing control (ECE 0.092 at $\alpha = 0.70$) and is explained by confidence shrinkage toward uniform.
- The 64 changed decisions on ViHSD are fully decomposable into rule-driven and threshold-driven causes, with all corrections rule-supported, providing auditable provenance for audit and appeal.

The contribution is not an accuracy breakthrough but a production-ready hybrid system that combines neural accuracy with rule-level explainability, supported by real-time latency on a single laptop GPU.

Disclosure of Interests. The authors have no competing interests to declare.

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